

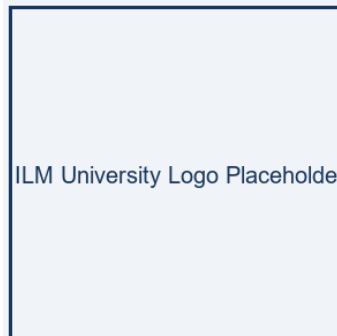
Report
of
**AI & AUTOMATION SYSTEMS FOR
CABLE
MANUFACTURING & OPERATIONS**
KDI Power Business Automation & AI Platform

Submitted To

Department of Computer Science and Engineering
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: **CS-2341412**
Name of Student: **GIRISH KUMAR YADAV**
Batch: **2023–27 (Section 4CSE4)**

31 August 2026

Candidate's Declaration

I, **GIRISH KUMAR YADAV** do hereby declare that the internship report titled "**AI & AUTOMATION SYSTEMS FOR CABLE MANUFACTURING & OPERATIONS --- KDI Power Business Automation & AI Platform**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: GIRISH KUMAR YADAV

Roll No.: CS-2341412

Date: 31 August 2026

Acknowledgement

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to express my deepest appreciation to **Mr. Vijay Gautam**, Administrative Manager, KDI Power Private Limited, New Delhi, for providing me with the opportunity to intern at the organization and for his mentorship throughout the internship period.

I am profoundly grateful to the entire **Engineering & Operations Team** at KDI Power for their continuous support and technical insights.

I extend my sincere thanks to **IILM University, Greater Noida**, and the **School of Computer Science and Engineering** for facilitating this internship opportunity.

Signature: _____

Student Name: GIRISH KUMAR YADAV

Roll No.: CS-2341412

Date: 31 August 2026

Internship Completion Certificate

KDI POWER PRIVATE LIMITED

ISO 9001:2015, ISO 14001:2015, ISO 45001:2018 Certified Manufacturer
New Delhi, India

CERTIFICATE OF INTERNSHIP COMPLETION

Certificate No.: KDIP/INT/2026/AI-014 | Date of Issue: 2nd September 2026

This is to certify that **Mr. GIRISH KUMAR YADAV** (Roll No. CS-2341412) has successfully completed a two-month internship in **Artificial Intelligence & Automation** with **KDI Power Private Limited**, from **29 June 2026 to 31 August 2026**. During this period, he demonstrated strong technical aptitude, initiative, and commitment, and independently designed, developed, and deployed the KDI Power WhatsApp Business Messaging System, PDF Editor, and LinkedIn Lead Generator.

Authorized Signatory: VIJAY GAUTAM, Administrative Manager, KDI Power Pvt. Ltd.

Chapter 1: Project Description

1.1 Introduction

The global industrial manufacturing vertical, particularly electrical wire and cable manufacturing, operates in a high-demand environment where operations, sales, and field teams must manage trade inquiries, technical datasheets, competitive tenders, and employee tasks efficiently.

KDI Power Private Limited commissioned the development of an integrated suite of Artificial Intelligence and Automation systems. During my industrial internship from **29 June 2026 to 31 August 2026**, I designed, built, and deployed five core software systems: (1) WhatsApp AI Assistant & Sales Dashboard; (2) KDI Lead Intelligence Engine; (3) LinkedIn Tender & Scheme Tracker; (4) AI PDF Reconstructor & Editor; and (5) Google Sheets Task Automation Bot.

1.2 Organization Profile

KDI Power Private Limited is an ISO 9001, ISO 14001, and ISO 45001 certified manufacturer of electrical wires and cables headquartered in New Delhi. Its core product portfolio includes Low Voltage (LT) Power Cables, XLPE Cables, Aerial Bunched Cables (ABC), Control Cables, House Wires, and Rubber Cables.

1.3 Problem Statement

1. Delayed Sales Turnaround on WhatsApp during off-hours.
2. High Cost of International B2B Lead Discovery for export markets.
3. Manual Competitor & Tender Tracking across news and social channels.
4. Uneditable Technical Datasheet Scans needing layout/formula extraction.
5. Operational Task Tracking Overhead with manual phone follow-ups.

1.4 Project Objectives

- Build automated WhatsApp Bot with Meta Cloud API and Supabase pgvector RAG.
- Create a 3-tier lead discovery pipeline with Google Maps Playwright scraping and Groq scoring.
- Implement LinkedIn CDP scraper and Mistral AI post summarizer.
- Construct hybrid React/FastAPI PDF Editor using MinerU and NVIDIA NIM VLM.
- Deploy Google Sheets API webhook on Render with GitHub Actions cron runners.

1.5 Technologies and Tools Used

Category	Technology / Tool	Purpose
Backend	Python 3.10+, FastAPI, Flask, Streamlit	Microservices, APIs, Scrapers
Frontend	React 18, Vite, Chart.js, Glassmorphic CSS	Editor canvas, Sales dashboard UI
Databases	Supabase PostgreSQL, pgvector, SQLite	Relational storage, RAG embeddings
AI / LLMs	Groq (Llama 3.3), Mistral AI, NVIDIA NIM	Conversations, summaries, VLM OCR
Automation	Playwright (Chromium/CDP), MinerU	Google Maps, LinkedIn CDP, PDF OCR

1.6 System Architecture Diagrams

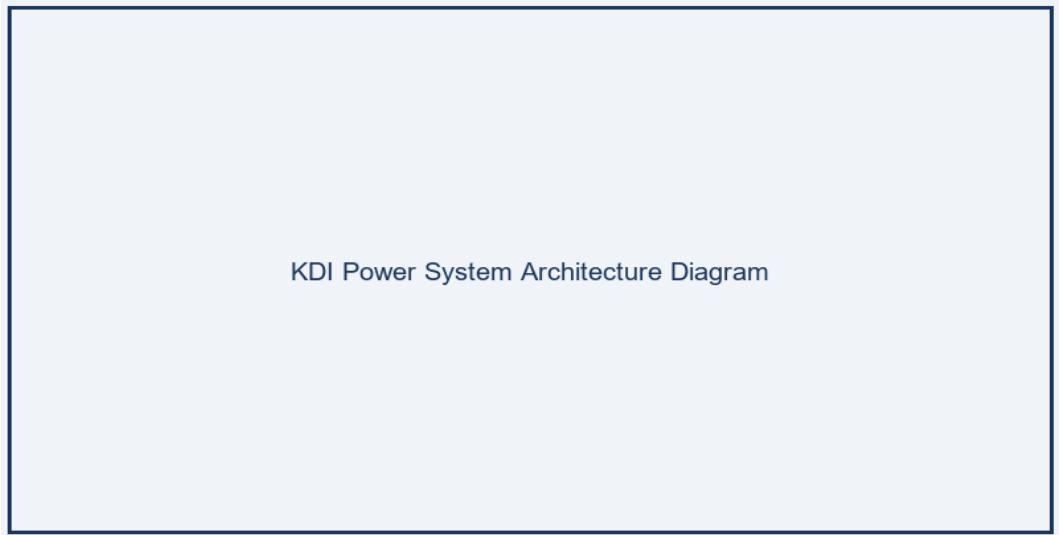


Figure 1.1: System Architecture Diagram of KDI Power Platform



Figure 1.2: End-to-End B2B Lead Intelligence Pipeline Workflow



Figure 1.3: Glassmorphic Sales Analytics Dashboard UI

Chapter 2: Bibliography / References

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Report Prepared By:

GIRISH KUMAR YADAV

AI & Automation Engineering Intern

KDI Power Private Limited, New Delhi

Date: 31 August 2026